

EPSON OPOS ADK MANUAL

**APPLICATION DEVELOPMENT
GUIDE**

POSPrinter (TM-U330x)

Version 2.80 Sep. 2015

Notes

- (1) Reproduction of any part of this documentation by any means is prohibited.
- (2) The contents of this documentation are subject to change without notice.
- (3) Comments and notification of any mistakes in this documentation are gratefully accepted.
- (4) This software cannot be used with other equipment than the specified.
- (5) EPSON will not be responsible for any consequences resulting from the use of any information in this documentation.

Trademarks

Microsoft®, Windows®, Windows Vista®, Windows Server®, Visual Basic® and Visual C++® are trademarks or registered trademarks of Microsoft Corporation in the United States and/or other countries.

EPSON® and ESC/POS® are registered trademarks of Seiko Epson Corporation.

Other product and company names used herein are for identification purposes only and may be trademarks or registered trademarks of their respective companies.

Contents

SECTION 1. INTRODUCTION	1
SECTION 2. DETAILS ON SETTINGS.....	2
2.1. References of Firmware Versions	2
2.2. Port Information	2
2.3. Device Settings	3
2.3.1. Usable Device Specific Settings.....	3
2.3.2. Options.....	3
SECTION 3. FUNCTION DETAILS	4
3.1. Property Set Values and Default Values.....	4
3.1.1. Capability Set Values.....	4
3.1.2. List Properties.....	6
3.1.3. Width and Height Properties	7
3.1.4. Common Property Strings	8
3.2. Methods.....	8
3.3. Escape Sequences	9
3.4. Power Condition Reports.....	10
3.5. Synchronous Processing	11
3.6. Printing Positions	11
3.7. Electronic Logo Function (NVRAM)	11
3.8. Printable Bitmap Types and Sizes.....	11
3.9. Maintenance Counter	12
3.10. Automatic Recovery Function	12
3.11. Output without Flow Control on the USB/Ethernet Interfaces	12
SECTION 4. WARNINGS.....	13

Section 1. Introduction

This manual describes the method of use and related items, as well as machine-specific precautions, when the EPSON TM-U330x Series POS Printers are used with the EPSON OPOS ADK program.

This manual applies to the following devices.

Device List

Serial	Parallel	USB	Ethernet
TM-U330BM	TM-U330BPM	TM-U330BMU	TM-U330BME
TM-U330BM_BM	TM-U330BPM_BM	TM-U330BM_BMU	TM-U330BM_BME
TM-U330DM	TM-U330DPM	TM-U330DMU	TM-U330DME
TM-U330DM_BM	TM-U330DPM_BM	TM-U330DM_BMU	TM-U330DM_BME

Before reading the manual, see the following explanation about the characteristic of the TM-U330x models.

- Station: Receipt (24-pin Serial impact dot matrix)
- 2-Color Printing
- Black Mark paper

Throughout the manual, the various model names will be referred to as TM-U330x. TM-U330x represents TM-U330BM, TM-U330BM_BM, TM-U330DM and TM-U330DM_BM.

Compatibility mode

The compatibility mode for upward compatibility was added in OPOS Ver2.60.

For the details of the compatibility mode, please refer to “EPSON OPOS ADK MANUAL APPLICATION DEVELOPMENT GUIDE Compatibility Mode”.

Section 2. Details on Settings

This section describes connection configurations and how to make the settings for the TM-U330x Series printers.

2.1. References of Firmware Versions

Refer to the release notes (Relnote.txt).

2.2. Port Information

1) Port information when serial port is used

The port information that can be set with the SetupPOS utility is as follows.

Item	Setting range
Baud rate [bps]	1200,2400,4800,9600,19200,38400,57600,115200
Bit length [bit]	8
Parity	NONE, ODD, EVEN
Stop bit [bit]	1
Handshake	DTR/DSR
Output buffer length [byte]	32 to 1024
Output interval time [ms]	0 to 9999

The default settings are as shown in the following table.

Item	Setting range
Baud rate [bps]	115200
Bit length [bit]	8
Parity	NONE
Stop bit [bit]	1
Handshake	DTR/DSR
Output buffer length [byte]	1024
Output interval time [ms]	2500

2) Port information when using parallel port

Not applicable

3) Port information when using USB port

Not applicable

4) Port information when using Ethernet port

Not applicable

2.3. Device Settings

The following explanation is about the settings for TM-U330x.

2.3.1. Usable Device Specific Settings

For the TM-U330x, the following device specific settings are settable by the SetupPOS utility. For the detail, please refer to the Section 2 of “EPSON OPOS ADK MANUAL APPLICATION DEVELOPMENT GUIDE POSPrinter (TM Series)”.

Tab	Settings
General	Disable panel buttons
	Assume print complete when data output finishes
	Homogenize Error Codes ^{*1}
	Ignore firmware version check
	Output complete timeout
Paper	Paper Type
	Paper Width [mm]: LineWidth [dot]: LineCharsList
Bitmap	TMFlogo...
Color Bitmap	Halftone: Method
	Halftone: Brightness
	Color: Primary
Status Log	ERROR
	OFFLINE
	Log file name (full path name)
	Maximum file size [KB]
Options	NearEndSensor
Printing Properties	Receipt Characters per Line
	Receipt Line Spacing [dots]
	CharacterSet [CodePage Number]

^{*1} The settings can be changed when using a connection other than serial.

2.3.2. Options

“NearEndSensor” check box

For the models, which are supported NearEndSensor function as an option, marking the check box could set NearEndSensor function.

Section 3. Function Details

This section describes the functions of the TM-U330x printers in details. Supplementary explanation of the parts not described in detail in the "UPOS" is also given here.

3.1. Property Set Values and Default Values

The following explanation is about the property set values and the default values.

3.1.1. Capability Set Values

The following values are the Capability set values.

Capability Name	Set Value
CapTransaction	TRUE
CapCoverSensor	TRUE
CapConcurrentRecSlp	FALSE
CapConcurrentJrnSlp	FALSE
CapConcurrentJrnRec	FALSE
CapConcurrentPageMode	FALSE
CapCharacterSet	PTR CCS KANJI
CapMapCharacterSet	FALSE
CapJrnUnderline	FALSE
CapJrnNearEndSensor	FALSE
CapJrnItalic	FALSE
CapJrnEmptySensor	FALSE
CapJrnDwideDhigh	FALSE
CapJrnDwide	FALSE
CapJrnDhigh	FALSE
CapJrnColor	0
CapJrnCartridgeSensor	0
CapJrnBold	FALSE
CapJrn2Color	FALSE
CapJrnPresent	FALSE
CapRecPageMode	FALSE
CapRecUnderline	TRUE
CapRecStamp	FALSE
CapRecRotate180	TRUE
CapRecRight90	FALSE
CapRecPapercut	TRUE (TM- U330BM) FALSE (TM-U330DM)

CapRecNearEndSensor	FALSE ^{*1}
CapRecMarkFeed	0 PTR_MF_TO_CUTTER ^{*2} PTR_MF_TO_TAKEUP ^{*3}
CapRecLeft90	FALSE
CapRecItalic	FALSE
CapRecEmptySensor	TRUE
CapRecDwideDhigh	TRUE
CapRecDwide	TRUE
CapRecDhigh	TRUE
CapRecColor	PTR_COLOR_PRIMARY PTR_COLOR_CUSTOM1
CapRecCartridgeSensor	0
CapRecBold	TRUE
CapRecBitmap	TRUE
CapRecBarCode	FALSE
CapRec2Color	TRUE
CapRecPresent	TRUE
CapRecRuledLine	FALSE
CapSlpUnderline	FALSE
CapSlpRotate180	FALSE
CapSlpRight90	FALSE
CapSlpNearEndSensor	FALSE
CapSlpLeft90	FALSE
CapSlpItalic	FALSE
CapSlpEmptySensor	FALSE
CapSlpDwideDhigh	FALSE
CapSlpDwide	FALSE
CapSlpDhigh	FALSE
CapSlpColor	0
CapSlpCartridgeSensor	0
CapSlpBothSidesPrint	FALSE
CapSlpBold	FALSE
CapSlpBitmap	FALSE
CapSlpBarCode	FALSE
CapSlp2Color	FALSE
CapSlpFullslip	FALSE
CapSlpPresent	FALSE
CapSlpPageMode	FALSE
CapSlpRuledLine	FALSE

^{*1} NearEndSensor is an option. When the mark the "NearEndSensor" check box at SetupPOS utility, this function is effective.

^{*2} This setting is valid only when TM-U330BM_BM is used.

^{*3} This setting is valid only when TM-U330DM_BM is used.

3.1.2. List Properties

The List Properties are explained in the following.

List Property	Settings
CharacterSetList	"254,255,437, 936"
JrnLineCharsList	""
RecLineCharsList (When 76.0 mm is set)	(Font A) "33" (Font B) "42"
RecLineCharsList (When 69.5 mm is set) ^{*1}	(Font A) "30" (Font B) "38"
RecLineCharsList (When 57.5 mm is set)	(Font A) "25" (Font B) "32"
SlpLineCharsList	""
RecBarcodeRotationList	""
RecBitmapRotationList	"0,180"
SlpBarcodeRotationList	""
SlpBitmapRotationList	""
FontTypefaceList	""

^{*1} The set values are valid only when width setting of TM-U330BM and TM-U330DM are 69.5mm.

3.1.3. Width and Height Properties

The width and height properties are described below.

Property	Settings		
	Default Value	Maximum value [dot]	Minimum value [dot]
RecLineSpacing	30	127	0
JrnLineSpacing	X	X	X
SlpLineSpacing	X	X	X
SlpLineHeight [dot]	X		
RecLineHeight [dot]	24		
JrnLineHeight [dot]	X		
SlpLineWidth [dot]	X		
RecLineWidth [dot] (When 76.0 mm is set)	450		
RecLineWidth [dot] (When 69.5 mm is set) *1	405		
RecLineWidth [dot] (When 57.5 mm is set)	337		
JrnLineWidth [dot]	X		
RecSidewaysMaxLines	0		
RecSidewaysMaxChars	0		
RecLinesToPaperCut*2	TM-U330B	TM-U330D	
	8	7	
SlpSidewaysMaxLines	X		
SlpSidewaysMaxChars	X		
SlpMaxLines	X		

X : No settings

^{*1} the set value is valid only when width setting of TM-U330BM and TM-U330DM are 69.5mm.

^{*2} It can be changed by the settings of the RecLineSpacing or the character height.

- When the RecLineSpacing is 0, the RecLinesToPaperCut is -1.

3.1.4. Common Property Strings

The Device information properties are described below.

- TM-U330BM

I/F	DeviceName	DeviceDescription
S	TM-U330BM	EPSON TM-U330BM POS Printer
P	TM-U330BPM	EPSON TM-U330BPM POS Printer
U	TM-U330BMU	EPSON TM-U330BMU POS Printer
E	TM-U330BME	EPSON TM-U330BME POS Printer

- TM-U330BM BM

I/F	DeviceName	DeviceDescription
S	TM-U330BM_BM	EPSON TM-U330BM_BM POS Printer
P	TM-U330BPM_BM	EPSON TM-U330BPM_BM POS Printer
U	TM-U330BM_BMU	EPSON TM-U330BM_BMU POS Printer
E	TM-U330BM_BME	EPSON TM-U330BM_BME POS Printer

- TM-U330DM

I/F	DeviceName	DeviceDescription
S	TM-U330DM	EPSON TM-U330DM POS Printer
P	TM-U330DPM	EPSON TM-U330DPM POS Printer
U	TM-U330DMU	EPSON TM-U330DMU POS Printer
E	TM-U330DME	EPSON TM-U330DME POS Printer

- TM-U330DM BM

I/F	DeviceName	DeviceDescription
S	TM-U330DM_BM	EPSON TM-U330DM_BM POS Printer
P	TM-U330DPM_BM	EPSON TM-U330DPM_BM POS Printer
U	TM-U330DM_BMU	EPSON TM-U330DM_BMU POS Printer
E	TM-U330DM_BME	EPSON TM-U330DM_BME POS Printer

I/F indicates the connected interface.

The following is the list of the four connecting interfaces.

S: Serial

P: Parallel

U: USB

E: Ethernet

3.2. Methods

The following explanation is about supported/unsupported Methods, and the

detailed information.

Method	Supported/Unsupported
PrintNormal	O
PrintTwoNormal	X
PrintImmediate	O
PrintBarCode	X
PrintBitmap	O
PrintMemoryBitmap	O
CutPaper ^{*1}	O (1~100: Cutting with one point of the bottom left corner uncut)
MarkFeed	X O ^{*2}
ChangePrintSide	X
ValidateData	O
TransactionPrint	O
SetLogo	O
SetBitmap	O
RotatePrint	O
EndRemoval	X
BeginRemoval	X
EndInsertion	X
BeginInsertion	X
ClearPrintArea	X
PageModePrint	X
DrawRuledLine	X

O: Supported

X : Unsupported

^{*1} The setting is available for the models other than TM-U330D.

^{*2} The MarkFeed method is supported by TM-U330BM_BM and TM-U330DM_BM only.

3.3. Escape Sequences

The following figure is about supported/unsupported Escape Sequences.

Escape Sequence	Support
#P ^{*1}	0~100
#fP ^{*1}	0~100
#sP	X
sL	X

#B	O
tL	O
bL	O
[*]#R	X
#IF	0~9999
#uF Base Pitch [inch]	0~ approx. 50 cm
#rF Maximum [inch]	2
[*]#E	0~65535
#fT	X
bC	O
!bC	O
#uC	1
iC	X
!iC	X
#rC	1~2
rvC	X
!rvC	X
#sC	X
#fC	X
tbC	X
!tbC	X
tpC	X
!tpC	X
1C	O
2C	O
3C	O
4C	O
#hC	1~2
#vC	1~2
cA	O
rA	O
!A	O
#stC	X
!stC	X
*#dL	X
N	O

O: Supported

X : Unsupported

Numbers: Settable range

*¹ TM-U330D is not supported. This is available only for the other models.

3.4. Power Condition Reports

The TM-U330x models support Power Condition Reports as follows.

Powered on reporting: Supported

Powered off reporting: Unsupported

3.5. Synchronous Processing

The TM-U330x models do not use Process ID to determine output completion.

3.6. Printing Positions

The TM-U330x models support the function for setting printing position.

Function	Support
Left margin	X
Printing Position	O

O: Supported

X : Unsupported

When the left margin setting function is supported, it is possible to specify the horizontal printing position of the bitmap or barcode by dots unit.

When the printing position settings are supported, it is possible to specify the horizontal printing position of the text, bitmap, or the barcode to the left, center, or the right side of the paper.

3.7. Electronic Logo Function (NVRAM)

The TM-U330x models feature a function for electronic logo. To use the electronic logo function (NVRAM), start "TMFlogo utility" from the "Device Specific Settings" of SetupPOS utility, and register image files (BMP style) at the function in advance. For the details of the registration, please refer to the "TM-U330 Utility User's Manual".

To print the registered image file, please use the following DirectIO.

PTR_DI_FLASH_BITMAP

For the details of the printing, please refer to Section 4 of "EPSON OPOS ADK APPLICATION DEVELOPMENT GUIDE POSPrinter (TM Series)".

3.8. Printable Bitmap Types and Sizes

The TM-U330x models support the following bitmap commands. For the detail, please refer to the Section 3 of "EPSON OPOS ADK MANUAL APPLICATION DEVELOPMENT GUIDE POSPrinter (TM Series)". The allowance ranges for bitmaps are as follows.

Bitmap command type	Allowance range
One-line bitmap	No setting range

3.9. Maintenance Counter

The TM-U330x models do not support the Maintenance Counter.

3.10. Automatic Recovery Function

The TM-U330x models feature a function for automatic recovery when the power is turned on again after an interruption of power. Recovery processing is performed automatically when the printer's power is turned on again after an interruption. The recovery processing restores the printer to the condition it was in before the power was interrupted.

The other models do not support this function.

3.11. Output without Flow Control on the USB/Ethernet Interfaces

The TM-U330x models do not support outputting without flow control on the USB/Ethernet interfaces.

Section 4. Warnings

This section describes precautions in use of TM-U330x.

There is no specific warning.